

Case interference and phrase length effects in processing Turkish center-embeddings



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- This study investigates doubly center-embedded structures with relative clauses (2-CE-RCs) in Turkish.
 - *E.g., The rat that the cat that the dog chased ate died.* (Chomsky & Miller, 1963, p. 286)
- 2-CE-RCs are reported to be extremely difficult to process due to (among many other factors)
 - nesting of the same type of clauses (e.g., Yngve, 1960)
 - the number of incomplete syntactic dependencies (Chomsky & Miller, 1963)
 - working memory limitations (Gibson, 1998; 2000)
 - similarity-based interference (Lewis & Vasishth, 2005)
 - non-optimal prosodic phrase lengths (Fodor, 2013)
- **Similarity-based interference:** easier processing of 2-CE-RCs when the similarity among the subject NPs is decreased
 - e.g., in terms of their case marking
- **Prosodic phrase lengths:** easier processing of 2-CE-RCs with **optimal** and **balanced** phrase lengths
 - NP1 || RC1 || VP1 in English
 - NP1 || NP6 || VP1 in Turkish

The Present Study



- Turkish, a highly inflectional and head-final language, can provide further data on similarity-based case interference and prosodic phrase lengths.
- An eye-tracking experiment was conducted to investigate:
 - if syntactic interference, phonological interference and/or prosodic phrase lengths influence the real-time processing of Turkish center-embeddings,
 - when (if any) the effects of syntactic interference, phonological interference and prosodic phrase lengths would reveal,
 - if there is an interaction between syntactic interference, phonological interference and prosodic phrase lengths.

- Experimental sentences were Turkish 2-CE-RCs nested as a complement clause inside a matrix clause:

Ø	[Marangoz-lar-ın	[nakliyeciler-in	[kiracı-nın	gri	koltuğ-u	yerleştirdiğ-i	oda-ya]
<i>Pro</i>	carpenter-PL-GEN	mover-PL-GEN	tenant-GEN	gray	sofa-ACC	place-PART-3SG	room-DAT
	NP1	NP2	NP3			VP3	
taşıdıkları	dolab-ı]	kurdukları-nı]	biliyorum.				
move-PART-3PL	cabinet-ACC	set.up-VN-3PL-ACC	know-PROG-1SG				
VP2		VP1					

'I know that the carpenters set up the cabinet that the movers moved to the room that the tenant placed the gray sofa in.'

- Experimental sentences were Turkish 2-CE-RCs nested as a complement clause inside a matrix clause:

Ø [Marangoz-lar-ın [nakliyeciler-in [kiracı-nın gri koltuğ-u yerleştirdiğ-i oda-ya]
Pro carpenter-PL-GEN mover-PL-GEN tenant-GEN gray sofa-ACC place-PART-3SG room-DAT
NP1 NP2 NP3 VP3
taşındıkları dolab-ı] kurdukları-nı] bil-iyor-um.
move-PART-3PL cabinet-ACC set.up-VN-3PL-ACC know-PROG-1SG
VP2 VP1

'I know that the carpenters set up the cabinet that the movers moved to the room that the tenant placed the gray sofa in.'

❓ Semantic support between the subjects and their verbs
to ease the processing of the already difficult structure
to prevent other plausible combinations of subjects and verbs

- Six conditions manipulating
 - Case interference: syntactic & phonological interference through case marking on the subject and object NPs
 - high syntactic – high phonological
 - low syntactic – high phonological
 - low syntactic – low phonological
 - Phrase lengths: (non-)optimal/(un-)balanced phrasing through phrase lengths
 - lengths encouraging optimal/balanced phrasing
 - lengths discouraging optimal/balanced phrasing
- 24 experimental sentences & 72 fillers, each followed by a comprehension question
- Task: read the sentences and answer the comprehension questions
- Participants: 43 native speakers of Turkish

Data analysis & Results



- [Marangozlar] [nakliyecilerin] [kiracının] oldukça geniş gri koltuğu büyük özenle yerleştirdiği odaya taşıdıkları dolabı kurdular sandım.
NP1 NP2 NP3 VP3 VP2 VP1 matrix V
↓ ↓
critical spill-over
region region

Overall accuracy > 70% ($M = 84\%$)

Critical region:

Discouraging > Encouraging

(t 's ≥ 3 , $z = 2.51$: first fixation, regr. path, rereading, total duration & pro. of regression out)

In both Discouraging and Encouraging Phrase Lengths Conditions:

HighSyn-HighPhono > LowSyn-HighPhono

(t 's ≥ 2.31 , gaze, regr. path, reread. and total duration)

HighSyn-HighPhono > LowSyn-LowPhono

(t 's ≥ 2.32 , $z = 2.63$: gaze, regr. path., reread., total duration & pro. of regr. out)

LowSynLowPhono = LowSynHighPhono

Spill-over region:

Discouraging > Encouraging

($t = 2.52$ in regression path dur.)

In both Discouraging and Encouraging Phrase Lengths Conditions:

HighSyn-HighPhono = LowSyn-HighPhono

HighSyn-HighPhono = LowSyn-LowPhono

LowSynLowPhono = LowSynHighPhono

Mixed effects linear/logistic regression models

- Prosodic phrase lengths & case interference as fixed effects
- Subjects & items as random effects

Summary & Conclusions

Syntactic interference:

- Decreasing syntactic interference eased the processing of Turkish CEs
- This was the case regardless of phonological interference

Phonological interference:

- Did not affect the processing difficulty of Turkish CEs. **WHY?**
 - the complex structure of experimental stimuli, i.e., 2-CE-RCs nested inside another clause (e.g., Nakayama et al., 2005)
 - no grammatically-permissible conditions available that controlled for syntactic int. while manipulating phonological int.
 - phonological interference may not cause similarity-based retrieval interference effects at all (e.g., Obata et al., 2010)

Prosodic phrase lengths:

- Prosodic phrase lengths encouraging the optimal phrasing of the sentence eased the processing of Turkish CEs
- Prosodic phrase lengths effects persisted to the spill-over region. **WHY?**
 - prosodic information may be used during reanalysis (Marcus & Hindle, 1990)
 - the final interpretations may be determined by phrase lengths
 - an end-of-sentence acceptability judgment task to investigate this possibility

THANK YOU!

Norming study for phrase lengths in Turkish

- A norming study was conducted to examine Turkish speakers' preferences for the prosodic phrasing of Turkish center-embeddings.
- 6 linguistically-informed judges were asked to insert two-three commas in the sentences.
- For the prosodic phrasing of the sentences as NP1 || NP6 || VP1;
 - all the judges placed a comma after NP1, in all the sentences (100%),
 - the number of commas after NP6 was 48 (33%),
 - two judges also placed a comma after NP2, in a total of 44 sentences (30.5%).
- There was no reliable difference between the sentences encouraging or discouraging the optimal phrasing.

Materials (Encouraging Phrase Lengths)

a. ENC, HighSyn-HighPhono

∅ [İşinin ehli marangoz-lar-in || [nakliyeciler-in [kiracı-nın gri koltuğ-u yerleştir-diğ-i] ...
Pro expert carpenter-PL-GEN mover-PL-GEN tenant-GEN gray sofa-ACC place-PART-3SG.POSS
NP1 NP2 NP3 VP3

b. ENC, LowSyn-HighPhono

∅ [İşinin ehli marangoz-lar-∅ || [nakliyeciler-in [kiracı-nın koltuğ-un minder-leri-ni yerleştir-diğ-i] ...
Pro expert carpenter-PL-NOM mover-PL-GEN tenant-GEN sofa-GEN cushion-3PL.POSS-ACC place-PART-3SG.POSS
NP1 NP2 NP3 NP4 VP3

c. ENC, LowSyn-LowPhono

∅ [İşinin ehli marangoz-lar-∅ || [nakliyeciler-in [kiracı-nın gri koltuğ-u yerleştir-diğ-i] ...
Pro expert carpenter-PL-NOM mover-PL-GEN tenant-GEN gray sofa-ACC place-PART-3SG.POSS
NP1 NP2 NP3 VP3

... oda-ya taşı-dık-ları] dolab-ı || dikkatli şekilde kur-duk-ları-nı/kur-du-lar] san-dı-m.
room-DAT move-PART-3PL.POSS closet-ACC carefully set.up-VN-3PL.POSS-ACC/set.up-PF-3PL think-PF-1SG
VP2 VP1

Followed by a content-neutral two-word follow-up sentence □ Yakında taşınacaklarmış.

'I thought that the expert carpenters carefully set up the closet that the movers moved to the room where the tenant placed the gray sofa. I heard they would move in soon.'

Materials (Discouraging Phrase Lengths)

d. DISC, HighSyn-HighPhono

∅ [Marangoz-lar-in] || [nakliyeciler-in] [kiracı-nin] oldukça geniş gri koltuğ-u büyük özen-le ...
Pro carpenter-PL-GEN mover-PL-GEN tenant-GEN extremely large gray sofa-ACC great care-with
NP1 NP2 NP3

e. DISC, LowSyn-HighPhono

∅ [Marangoz-lar-∅] || [nakliyeciler-in] [kiracı-nin] koltuğ-un oldukça geniş minder-leri-ni büyük özen-le ...
Pro carpenter-PL-NOM mover-PL-GEN tenant-GEN sofa-GEN extremely large cushion-3PL.POSS-ACC great care-with
NP1 NP2 NP3 NP4

f. DISC, LowSyn-LowPhono

∅ [Marangoz-lar-∅] || [nakliyeciler-in] [kiracı-nin] oldukça geniş gri koltuğ-u büyük özen-le ...
Pro carpenter-PL-NOM mover-PL-GEN tenant-GEN extremely large gray sofa-ACC great care-with
NP1 NP2 NP3

... yerleştir-diğ-i] oda-ya taşı-dık-ları] dolab-ı || kur-duk-ları-nı/kur-du-lar] san-dı-m...
place-PART-3SG.POSS room-DAT move-PART-3PL.POSS closet-ACC set.up-VN-3PL.POSS-ACC/set.up-PF-3PL think-PF-1SG
VP3 VP2 VP1

Followed by a content-neutral two-word follow-up sentence □ Yakında taşınacaklarmış.

'I thought that the carpenters set up the closet that the movers moved to the room where the tenant placed the extremely large gray sofa with great care. I heard they would move in soon.'

Data Analysis

Six eye-tracking measures:

Rayner et al., 1989)

- First fixation duration (FD)
- Gaze duration (GD)
- Regression path duration (RPD)
- Rereading duration (RRD)
- Total duration (TD)
- Probability of regression out (PRO)



Early processes of comprehension



Later processes of comprehension

- **Two sets of analyses:**

1st set: Prosodic Phrase Lengths & Case Interference (comparison of HighSyn–HighPhono and LowSyn–HighPhono to LowSyn–LowPhono)

2nd set: Prosodic Phrase Lengths & Case Interference (comparison of LowSyn–HighPhono to HighSyn–HighPhono)

Mixed effects linear regression models (for FD, GD, RPD, RRD and TD) or logistic regression models (for PRO)

(Bates et al., 2015)

- Prosodic phrase lengths & case interference as fixed effects
- Subjects & items as random effects
- Random slopes for subjects and/or items by the predictor variables when the model allowed

(Barr et al., 2013)

R packages

- *lme4* (Bates & Maechler, 2013)
- *languageR* (Baayen, 2013)
- *influence.ME* (Nieuwenhuis et al., 2012)

Descriptive Results (Case Interference)

Critical Region

	FFD	GD	RPD	RRD	TD	PRO
HighSyn-HighPhono, ENC	231 (6)	380 (14)	428 (19)	315 (31)	661 (31)	.10 (.02)
LowSyn-HighPhono, ENC	233 (5)	318 (12)	363 (17)	253 (26)	565 (28)	.11 (.02)
LowSyn-LowPhono, ENC	238 (7)	333 (12)	418 (24)	275 (30)	602 (30)	.15 (.03)
HighSyn-HighPhono, DISC	239 (6)	409 (16)	554 (27)	434 (36)	793 (35)	.20 (.03)
LowSyn-HighPhono, DISC	247 (8)	347 (14)	462 (23)	327 (32)	633 (31)	.21 (.03)
LowSyn-LowPhono, DISC	239 (6)	324 (13)	417 (21)	352 (32)	648 (31)	.19 (.03)

Spill-over Region

	FFD	GD	RPD	RRD	TD	PRO
HighSyn-HighPhono, ENC	244 (7)	283 (10)	373 (27)	146 (18)	408 (19)	.15 (.03)
LowSyn-HighPhono, ENC	238 (6)	267 (9)	332 (18)	180 (23)	439 (24)	.19 (.03)
LowSyn-LowPhono, ENC	239 (7)	290 (11)	378 (24)	178 (26)	447 (25)	.19 (.04)
HighSyn-HighPhono, DISC	247 (7)	280 (19)	434 (26)	210 (27)	463 (24)	.26 (.04)
LowSyn-HighPhono, DISC	249 (7)	297 (11)	416 (27)	170 (22)	466 (26)	.26 (.04)
LowSyn-LowPhono, DISC	251 (8)	278 (11)	416 (27)	192 (22)	465 (22)	.24 (.04)

Descriptive Results (Phrase Lengths)

Critical Region

	FFD	GD	RPD	RRD	TD	PRO
HighSyn-HighPhono, ENC	231 (6)	380 (14)	428 (19)	315 (31)	661 (31)	.10 (.02)
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